UNIT 7 · STANDARD 6.EE.7

Equations and Inequalities Problem Solving

Lesson 7-7

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

'Twice a number is 18' → $2n = 18$ — the equation models the words

Model

To show a real-life situation with an equation or inequality.

$x + 5 = 12$ — exactly one answer: $x = 7$

Equation

A math sentence with an equal sign showing both sides are the same.

$x + 5 > 12$ — many answers: $x = 8, 9, 10, \ldots$

Inequality

A math sentence that compares two sides with <, >, ≤, or ≥.

If the problem asks for a number of people and you get $x = -3$, that is NOT reasonable

Reasonableness

Checking if your answer makes sense.

"5 less than a number is 12" becomes $x - 5 = 12$, where x is the variable.

Variable

A letter that stands for the unknown amount in a word problem.

"You can spend at most \$20" is a constraint: cost ≤ 20.

Constraint

A limit in a problem that tells what values are allowed.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can model and solve real-world problems using equations and inequalities.

1️⃣ Notice

Detective Santos is closing a complex case.

2️⃣ Key idea

I can model and solve real-world problems using equations and inequalities.

3️⃣ Words I'll use

Model **Equation** **Inequality** **Reasonableness** **Variable** **Constraint**

🤔 Think About It

- Which clue leads to an equation and which leads to an inequality?
- What operation does 'divided equally among 4' suggest?
- What does 'less than $50' tell you about the inequality symbol?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: A museum has some paintings. After adding 12, they have 45. Which equation models this?

A. $p + 12 = 45$

B. $p - 12 = 45$

C. $12p = 45$

D. $p / 12 = 45$

Step 1 'Adding 12' to the original number gives 45: $p + 12 = 45$.

Step 2 Solve: $p = 33$.

✅ **Answer:** A. $p + 12 = 45$

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A detective needs more than 8 hours to finish the investigation. She has already worked 3 hours. Which inequality represents the additional hours h she needs?

A. $3 + h > 8$

B. $3 + h = 8$

C. $h > 3$

D. $8 + h > 3$

Step 1 ______

Step 2 ______

Answer: ______

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which model is correct for: 'A number divided by 6 equals 7'?

A. $n / 6 = 7$

B. $6n = 7$

C. $n - 6 = 7$

D. $n + 6 = 7$

Show your work:

______

______

______

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Before solving a word problem, how do you decide whether to model it with an equation or an inequality?

Level 1 support

Start here: An exact total points to an equation; a limit or range points to an inequality.

💡 **Need a hint?**

Sentence starters (optional)

I would model this with a(n) ____ because the problem says ____.

Modelaría esto con un(a) ____ porque el problema dice ____.

Words like ____ tell me to use ____.

Palabras como ____ me dicen que use ____.

Word bank: model | equation | inequality | reasonableness

Listen for: *Students use signal words ('equals/total' for an equation; 'at least/at most/no more than' for an inequality) to choose the correct model.*

Level 2

How would the same situation change if 'spends exactly \$50' became 'spends at most \$50'? Justify which model fits each version.

- 'Exactly \$50' needs a(n) ____, but 'at most \$50' needs a(n) ____ because ____.
- The model changes because ____.

EXPLORE

What was your plan to solve this problem, and how did you check that your answer was reasonable?

Level 1 support

Start here: I model the problem, solve it, then check the answer against the situation.

💡 **Need a hint?**

Sentence starters (optional)

My plan was to ___, and I knew to ___ because ___.

Mi plan fue ___, y supe ___ porque ___.

My answer is reasonable because ___.

Mi respuesta es razonable porque ___.

Word bank: **model** **equation** **inequality** **reasonableness**

Listen for: *Students describe modeling the situation, solving with inverse operations, and judging reasonableness by checking the answer fits the real context.*

Level 2

Suppose your solution is a negative number or a fraction that doesn't fit the situation. What should you do, and why does reasonableness matter?

- If my answer doesn't fit, I should ___ because ___.
- Reasonableness matters because ___.

CONNECT

Describe a real situation that an equation or inequality could model, and explain which one you would use.

Level 1 support

Start here: Real situations with limits or exact totals can be modeled mathematically.

💡 **Need a hint?**

Sentence starters (optional)

A real situation is ____, which I would model with a(n) ____.

Una situación real es ____, que modelaría con un(a) ____.

I chose this model because the situation has ____.

Elegí este modelo porque la situación tiene ____.

Word bank: model | equation | inequality | reasonableness

Listen for: *Students give a realistic context and correctly match it to an equation (exact value) or inequality (range/limit), justifying their choice.*

Level 2

Critique: 'Any word problem can be solved with an equation, so inequalities are never needed.' Use an example to argue why inequalities are sometimes the only correct model.

- Inequalities are needed when ____.
- An example that cannot use a single equation is ____ because ____.

Write About the Math

The Writing Revolution

I can explain my reasoning using the words model, equation, inequality, and reasonableness.

1. Kernel Sentence

subject + verb

Model: Equation is a math sentence with an equal sign showing both sides are the same.
Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.

Write a kernel sentence about equation. Use a subject and a verb.
Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Equation matters in math
Ecuación importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Equation matters in math because ___.** *Ecuación importa en matemáticas porque ___.*
but *pero*	**Equation matters in math, but ___.** *Ecuación importa en matemáticas, pero ___.*
so *entonces*	**Equation matters in math, so ___.** *Ecuación importa en matemáticas, entonces ___.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equation.
Di un hecho verdadero sobre equation.

Equation ____.

Question
Pregunta

Ask a question about equation.
Haz una pregunta sobre equation.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equation.
Muestra entusiasmo sobre equation.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equation.
Dile a un compañero qué hacer con equation.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

I would model this with a(n) ____ because the problem says ____.
Modelaría esto con un(a) ____ porque el problema dice ____.

Words like ____ tell me to use ____.
Palabras como ____ me dicen que use ____.

My plan was to ____, and I knew to ____ because ____.
Mi plan fue ____, y supe ____ porque ____.

Try It

Solve on your own. Check the answer key when you are done.

1. You have $50 and spend $8 per ticket. Which inequality shows how many tickets t you can buy?

A. $8t \le 50$

B. $8t \ge 50$

C. $t + 8 \le 50$

D. $8t = 50$

Show your work:

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2. Solve: $x - 12 = 20$.

A. $x = 32$

B. $x = 8$

C. $x = 240$

D. $x = -8$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Create two word problems about the same situation: one that requires an equation and one that requires an inequality. Solve both and explain why the models are different.

Sentence starter: Equation problem: ____. Model: ____. Solution: ____. Inequality problem: ____. Model: ____. Solution: ____. The models are different because ____.

Show your work:

Reflect — Exit Ticket

A box of donuts has some donuts. After giving away 7, there are fewer than 5 left. Which inequality represents the starting number of donuts d?

A. $d - 7 < 5$

B. $d + 7 < 5$

C. $d - 7 > 5$

D. $d - 7 = 5$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $3 + h > 8$ *— 3 hours plus additional hours h must be more than 8: $3 + h > 8$. Solve: $h > 5$.*
2. **You Try (notes):** A. $n / 6 = 7$ *— Divided by 6 is $n / 6$. Equals 7 means = 7. So $n / 6 = 7$.*
3. **Try It 1:** A. $8t \le 50$ *— Total spent must be 50 or less: $8t \le 50$ (so $t \le 6$ whole tickets).*
4. **Try It 2:** A. $x = 32$ *— Add 12 to both sides: $x = 32$.*
5. **Exit Ticket:** A. $d - 7 < 5$ *— Starting with d donuts and giving away 7 leaves fewer than 5: $d - 7 < 5$. Solve: $d < 12$. ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).